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Evolutionary multiobjective optimization in water resources: The past, present, and future

P.M. Reed [a,*], D. Hadka [b], J.D. Herman [a], J.R. Kasprzyk [a], J.B. Kollat [a]

[a] Department of Civil and Environmental Engineering, The Pennsylvania State University, University Park, Pennsylvania, USA
[b] Department of Computer Science and Engineering, The Pennsylvania State University, University Park, Pennsylvania, USA

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ABSTRACT

This study contributes a rigorous diagnostic assessment of state-of-the-art multiobjective evolutionary algorithms (MOEAs) and highlights key advances that the water resources field can exploit to better discover the critical tradeoffs constraining our systems. This study provides the most comprehensive diagnostic assessment of MOEAs for water resources to date, exploiting more than 100,000 MOEA runs and trillions of design evaluations. The diagnostic assessment measures the effectiveness, efficiency, reliability, and controllability of ten benchmark MOEAs for a representative suite of water resources applications addressing rainfall–runoff calibration, long-term groundwater monitoring (LTM), and risk-based water supply portfolio planning. The suite of problems encompasses a range of challenging problem properties including (1) many-objective formulations with four or more objectives, (2) multi-modality (or false optima), (3) nonlinearity, (4) discreteness, (5) severe constraints, (6) stochastic objectives, and (7) non-separability (also called epistasis). The applications are representative of the dominant problem classes that have shaped the history of MOEAs in water resources and that will be dominant foci in the future. Recommendations are given for the new algorithms that should serve as the benchmarks for innovations in the water resources literature. The future of MOEAs in water resources needs to emphasize self-adaptive search, new technologies for visualizing tradeoffs, and the next generation of computing technologies.

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1. Introduction

The multiobjective nature of water resources challenges is clearly evident in the vast literature that has emerged in the areas of observation networks, model calibration, and water resources planning and management. Haimes and Hall [1] encapsulate the *many-objective* nature of water resources in their contribution to the inaugural volume of this journal. Their study has tremendous foresight and frames the need to address water resources challenges using multiobjective analyses that consider objectives for cost, robustness, adaptivity, stability, and irreversibility. They warn that modern tools and frameworks must consider "... uncertainties and errors in systems modeling... associated with the following model characteristics: model structure (topology), model parameters, model scope or focus, data, optimization technique, and human subjectivity".

Three decades later, the motivating drivers and concerns elucidated by Haimes and Hall [1] are prominently shaping state-of-the-art multiobjective modeling efforts. Abstracting their concerns, water resources problems are often characterized by several of the following mathematical properties: nonlinearity, stochasticity, discreteness, non-convexity, high dimension decisions, severe combinatorial growth rates, and uncertainty. It is these very properties that have motivated the past 20 years of research advancing the effective use of multiobjective evolutionary algorithms (MOEAs) for discovering and exploiting the critical tradeoffs facing water resources systems. As defined by Coello Coello et al. [2], MOEAs can be classified as *a posteriori* decision support tools. This means that decision makers are presented with an explicit representation of their tradeoffs before they express their preference for one or more selected solutions.

In their 1975 review of multiobjective optimization in water resources, Cohon and Marks [3] framed the multiobjective design problem as "vector optimization" subject to a set of constraints defining the feasible region of the decision space. Eq. (1) formally defines the multiobjective design problem as minimization of a vector, $F(\mathbf{x})$, composed of M objective functions, $f(\mathbf{x})$

$$\begin{aligned} \underset{\mathbf{x}\in\Omega}{\text{minimize}} \quad & F(\mathbf{x}) = [f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_M(\mathbf{x})] \\ \text{subject to} \quad & c_m(\mathbf{x}) = 0, \forall m \in \mathcal{E}, \\ & c_n(\mathbf{x}) \leqslant 0, \quad \forall n \in \mathcal{I}. \end{aligned} \tag{1}$$

* Corresponding author.
E-mail addresses: preed@engr.psu.edu (P.M. Reed), dmh309@psu.edu (D. Hadka), jdh33@psu.edu (J.D. Herman), jrk301@psu.edu (J.R. Kasprzyk), juk124@psu.edu (J.B. Kollat).

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Notation index

Mathematical notation

Ω	total decision space
Λ	feasible decision space
L	number of decision variables
M	number of objectives
$\mathbf{x}$, $\mathbf{y}$, $\mathbf{z}$	decision vectors
x_i	the ith variable in a decision vector
F	the multiobjective function
f_i	the ith objective function
$\mathcal{E}$	the set of indices of equality constraints
$\mathcal{I}$	the set of indices of inequality constraints
c_i	the ith constraint function
I_G	generational distance indicator
I_H	hypervolume indicator
I_ϵ	additive ϵ-indicator
$\mathcal{PF}$	local Pareto approximate set
$\mathcal{PF}'$	reference (or best known) Pareto approximate set
d	Euclidean distance function
α	attainment function for hypervolume calculation

Acronyms

DE	differential evolution
EA	evolutionary algorithm
LHS	latin hypercube samples
MO	multi-objective
MOEA	multi-objective evolutionary algorithm
NFE	number of function evaluations
PCX	parent-centric crossover
PM	polynomial mutation
PSO	particle swarm optimization
SBX	simulated binary crossover
SPX	simplex crossover
UM	uniform mutation
UNDX	unimodel normal distribution crossover

Algorithms

AMALGAM	a multi-algorithm genetically adaptive method
Borg	auto-adaptive multi-operator MOEA
GDE3	generalized differential evolution 3
IBEA	indicator based evolutionary algorithm
MOEA/D	MOEA based on decomposition
NSGAII	nondominated sorting genetic algorithm II
OMOPSO	multiobjective PSO algorithm proposed by [65]
SPEA2	strength-pareto evolutionary algorithm 2
ε-MOEA	ε-dominance MOEA
ε-NSGAII	ε-dominance NSGAII

where $\mathbf{x} = [x_1, x_2, \ldots, x_L]$ (a vector of L decision variables in the decision space, Ω). The decision variables can be real-valued, integers, or binary. Eq. (1) optionally subjects $F(\mathbf{x})$ to one or more *equality* and/or *inequality* contraints defined by $\mathcal{E}$ and $\mathcal{I}$, respectively. The *feasible region* Λ is then the set of decisions in Ω that satisfy these constraints.

Cohon and Marks [3] noted that in the absence of additional preference information, there is no single optimal solution to the multiobjective design problem when conflicts exist between one of more of the objective functions. Conflicts exist when improvements in one objective can only be accomplished by degrading performance in one or more of the $M-1$ remaining objectives. Consequently, an optimal set of non-inferior decision vectors that represent the conflicts or tradeoffs between the various objective functions compose what is termed the non-dominated or Pareto-optimal set [4].

Assuming minimization of all objectives, an objective vector $F(\mathbf{x}) = [f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_M(\mathbf{x})]$ Pareto dominates another objective vector $F(\mathbf{y}) = [f_1(\mathbf{y}), f_2(\mathbf{y}), \ldots, f_M(\mathbf{y})]$, denoted by $F(\mathbf{x}) \prec F(\mathbf{y})$, if and only if $\forall i \in \{1,2,\ldots,M\}$, $f_i(\mathbf{x}) \leqslant f_i(\mathbf{y})$, and there exists at least one $j \in \{1,2,\ldots,M\}$ where $f_j(\mathbf{x}) < f_j(\mathbf{y})$. This can be more simply stated that a non-dominated or Pareto-optimal solution's performance is not exceeded with respect to all objectives by any other solution. The *Pareto optimal set* is then defined as the set of all solutions that are non-dominated with respect to one another and also satisfy any feasibility constraints. For non-trivial applications, it is difficult or computationally intractable to enumerate all possible solutions. Therefore in most water resources applications, the solution set found by MOEAs is typically considered the best known approximation to the Pareto set (i.e., the Pareto approximate set). A plot of the objective values for the Pareto optimal set is generally referred to as the Pareto frontier. High quality approximations of the Pareto frontier distinguish how a single objective cannot be improved without degrading performance in one or more of the remaining objectives (i.e., diminishing returns in the *a posteriori* decision support context).

Goldberg [5] was the first to note that the population-based search of evolutionary algorithms could be exploited to approximate the Pareto frontier for an application using a single optimization run. His suggestion was to optimize each solution's *non-dominance* until the algorithm's population approaches the globally non-dominated set of alternatives (i.e., the *Pareto optimal set*). The early MOEA studies exploiting Goldberg's [5] suggestion formed the initial generation of MOEA tools [6–9]. The link between early MOEA efforts and water resources applications emerged very quickly as a result of interactions between David Goldberg's and Wayland Eheart's research groups at the University of Illinois [10,11]. This early work focused on the monitoring and management of contaminated groundwater resources. Impressively, Meyer et al. [12] provided one of the first attempts to use heuristic optimization to approximate the discrete, nonlinear, and uncertain *many-objective* (i.e., optimizing four or more objectives) tradeoffs for a groundwater monitoring application. A few years later, Hoshin Gupta's group further motivated the use of MOEAs to address calibration of conceptual rainfall–runoff models [13,14]. As reviewed by Nicklow et al. [15], these early studies have helped MOEA applications to have a dominant influence on a broad range of water resources application areas.

The purpose of this paper is to provide a rigorous diagnostic assessment of state-of-the-art MOEAs and highlight key advances that the water resources area can exploit to better address the challenges highlighted by Haimes and Hall [1] in future water resources research. Our goal in this paper is not to summarize the entire body of work within the water resources domain, but to focus on promising issues and tools that have not yet been widely adopted in the water resources area. Building on Hadka and Reed [16], this study provides the most comprehensive diagnostic assessment of MOEAs for water resources that has ever been attempted. We contribute a careful assessment of the effectiveness, efficiency, reliability, and controllability of ten benchmark MOEAs for a representative suite of water resources applications addressing rainfall–runoff model calibration, long-term groundwater monitoring (LTM), and risk-based water supply portfolio planning. The applications are representative of dominant problem classes that have shaped the history of MOEAs in water resources and that will be dominant foci in the future. The suite of problems

selected for our diagnostic assessment of MOEAs encompass a range of challenging problem properties including (1) many-objective formulations (four or more objectives), (2) multi-modality (or false optima), (3) nonlinearity, (4) discreteness, (5) severe constraints, (6) stochastic objectives, and (7) non-separability (also called epistasis). Readers interested in prior reviews of MOEA research within and outside of the water resources area can refer to [2,15–20].

Readers interested in discussions of the difficulties associated with classical multiobjective programming, a basic background on evolutionary computation, or MOEAs can reference [20,21,2].

2. Important innovations

2.1. Conditions for a proof of convergence and diversity

The first generation MOEAs discussed in Section 1 were in fact strongly limited in their ability to sustain convergence (evolving to the Pareto front) while maintaining solution diversity (distributing solutions on the full extent of the Pareto front). The early algorithms largely failed to solve non-trivial problems [2]. The properties of convergence and diversity are critical to the success of any MOEA. The Non-dominated Sorting Genetic Algorithm-II (NSGAII) [22] can be credited as the primary technological breakthrough that transitioned MOEAs into wide application by adding elitism, efficient non-domination sorting, and parameter free diversity maintenance. These innovations dramatically improved the range of multiobjective problems where convergence and diversity could be sustained. Subsequent to the NSGAII, a new generation of MOEAs has emerged that has had a large impact in water resources and beyond (for a detailed history and review see Coello Coello et al. [2]). Although modern MOEAs have dramatically increased the size and scope of multiobjective problems that can be addressed, many of them fail to satisfy the theoretical conditions required for proofs of convergence and diversity maintenance.

Rudolph and Agapie [23,24] as well as Laumanns et al. [25] provide a theoretical foundation for the requisite properties that a MOEA must satisfy to have a proof of convergence and diversity (i.e., given unlimited search time the algorithm has a non-zero probability of generating a well-spread set of Pareto optimal solutions). Our intention in this text is not to claim that an MOEA must have a proof of convergence to have practical merit. Rather, algorithms with a sound theoretical foundation for preserving convergence and diversity will be important for maximizing the value of MOEAs in challenging water resources applications. A first condition for a proof of convergence requires that recombination operators have "diagonal-positive transition matrices" [24]. Put more simply, modern MOEAs exploiting tournament selection with replacement and that couple recombination operators with a form of mutation in which the full decision space is accessible have diagonal-positive transition matrices.

As shown by Rudolph [23] the second requisite condition for an MOEA's proof of convergence is "elite preservation". Elite preservation means that as evolution progresses, non-dominated solutions will not be lost in subsequent generations (i.e., they avoid the failure mode termed *deterioration*). Archiving of these high-quality solutions is the primary mechanism for avoiding deterioration failures, but as the number of objectives increases, it is vital to avoid unbounded growth of the stored non-dominated solutions in the archive. Laumanns et al. [25] shows that ε-dominance archiving satisfies elite preservation and guarantees a bounded archive size that varies with the decision maker's precision goals for each objective. The ε-dominance concept replaces point-based non-domination sorting with a grid-based sort where only a single solution is allowed to reside in each grid block. Kollat and Reed [26] provide a detailed illustration of ε-dominance and its ability to dramatically improve the computational scaling of MOEA search while preserving a diverse representation of the tradeoffs for a four-objective LTM application. Laumanns et al. [25] also proved that ε-dominance is guaranteed to preserve diversity during evolutionary search.

As a synopsis, when evaluating MOEAs it important to carefully consider their selection, recombination, mutation, and archiving strategies relative to the conditions discussed above. In subsequent sections, algorithms with theoretical proofs of convergence and diversity will be highlighted.

2.2. Many-objective optimization: overcoming biases and search failures

The term *many-objective optimization* is a reference to the rapidly growing trend in the MOEA field to consider four or more objectives simultaneously [16,18,27–44]. This body of literature addresses two major foci: (i) the value of discovering high dimensional tradeoffs and (ii) the algorithmic challenges limiting MOEA effectiveness.

Brill et al. [45] highlight that for complex planning problems, many-objective representations of tradeoffs can help reduce the negative impacts from two forms of decision bias. The first bias, *cognitive myopia* [46], occurs in low dimensional optimization when stakeholders inadvertently ignore alternatives that could strongly influence their decision preferences (i.e., not knowing what's possible). The second bias, *cognitive hysteresis* [47], refers to the challenge that lower dimensional, highly constrained problem representations often reinforce decision makers' preconceptions (i.e., not being able to discover new hypotheses). Kasprzyk et al. [39] provide a clear example of how many-objective optimization can be used to overcome these decision biases in the context of managing population and drought risks to urban water supply. There is a tremendous potential to innovate decision support processes and risk management in the water resources area with many-objective search and visualization technologies.

The second major focus of many-objective optimization has focused on the challenges and limitations of MOEAs in attaining high-dimensional tradeoff surfaces. The first factor impacting the difficulty of a problem is the actual dimension (or number) of objectives where legitimate conflicts exist [33,37]. For a problem with M objectives the theoretical upper bound for the number of conflicts is $M-1$, which corresponds to the dimension of the Pareto front. As noted by Brockhoff et al. [33], adding additional objectives can both increase or decrease the difficulty of a problem. Often in highly constrained instances, the addition of objectives provides MOEAs with more potential points to explore and can make a problem easier to solve. Alternatively, non-domination ranking becomes less discriminative and more challenging with an increasing number of objectives [48,31,36]. As non-domination sorting becomes less discriminative, it can lead to a failure mode termed *dominance resistance*, in which the rapid growth of locally non-dominated solutions overwhelms the ability of MOEAs' selection operators to distinguish high quality solutions [36]. This specific issue was one of the motivating factors for the theoretical work by Laumanns et al. [25] and their proposal of the ε-dominance approximation scheme (recall the discussion in Section 2.1). In this study, several of the modern algorithms that will be compared implement the ε-dominance approximation scheme as part of their search and/or archiving schemes to avoid deterioration and maintain highly scalable search.

2.3. Visual analytics: watching search and decision support

Understanding MOEA search dynamics in terms of both convergence and diversity is a challenging issue. Moreover, exploiting many-objective solution sets to facilitate design discoveries and improve hypotheses on system performance (i.e., to avoid cognitive myopia and hysteresis) requires highly interactive tools for exploring tradeoffs and their design consequences. There is a growing body of literature addressing these challenges termed *many-objective visual design analytics* [29,39,41–43,49–51]. The term visual analytics originates from *Illuminating the Path*, the vision provided by Thomas and Cook [52] for the future of using scientific visualization to advance decision making in complex systems. Their vision calls for a new generation of scientific visualization tools that facilitate the rapid analysis of large data sets using highly interactive software that enables multiple connected views of challenging planning problems. In the context of using MOEAs, users can "brush" away solutions in the objective tradeoffs that do not meet their performance threshold, while simultaneously visualizing their component decisions or designs. This research blends innovations in scientific visualization, human–computer interaction, and the cognitive sciences [49,53–55]. For future MOEA research, the ability to "watch" the evolution of many-objective Pareto approximate sets (i.e., visualize the Pareto-approximate set at each iteration of the evolution) as well as exploit them in decision making is critical. Moreover, there are new studies emphasizing the importance of interactively incorporating human expertise into the search process [56–60], even if it is something as simple as judging that additional search is not warranted given recent limited gains in convergence and/or diversity.

3. Modern multiobjective evolutionary algorithm suite

To assess the current state-of-the-art in MOEA performance, we selected a representative sample of modern MOEAs to test in this study. This section highlights the key characteristics of each MOEA. Readers interested in detailed descriptions of the algorithms should reference their source publications and authors' websites.

3.1. Historical benchmarks (NSGAII, SPEA2)

The NSGAII [22] is the classic example of a Pareto-based MOEA. The NSGAII and its ancestor, the NSGA, were the first MOEAs to use the Pareto dominance relation to successfully search for the entire Pareto front in a single run. While the introduction of the NSGAII revolutionized the field and spawned the creation of many sophisticated MOEAs, the NSGAII remains the most popular MOEA in active use today. The SPEA2 [61] is another popular MOEA that was introduced shortly after the NSGAII, but differs significantly in its use of Pareto dominance information. Unlike the NSGAII, which measures the fitness of solutions using pairwise Pareto dominance comparisons, the SPEA2 assigns each solution a fitness value based on the number of competing solutions it dominates. This measure is termed the strength of the solution. Given their sustained popularity in the literature, the NSGAII and the SPEA2 are included as baseline algorithms to compare against more recent contributions.

3.2. Pareto front approximation (ε-MOEA, ε-NSGAII, OMOPSO)

Hanne [62] observed that the replacement strategy used by many MOEAs suffers from the problem of deterioration. Deterioration is particularly prominent in MOEAs with fixed-sized populations, where the replacement strategy determines which offspring replace existing members in the population. If a population member is replaced by an inferior solution, then the quality of the approximation set deteriorates. Laumanns et al. [25] introduced the ε-dominance relation (and subsequently the ε-dominance archive) to eliminate the problem of deterioration. Rather than maintain a fixed-sized population, the size of an ε-dominance archive is dynamic, growing and shrinking as needed to maintain an approximate representation of the Pareto front, where ε defines the resolution of the approximation. The end result is an algorithm that guarantees both convergence and diversity [25]. The ε-MOEA, the ε-NSGAII and the OMOPSO are three modern MOEAs that use ε-dominance, but each differs significantly in their implementation.

The ε-MOEA [63] was the first MOEA to use the ε-dominance relation. The ε-MOEA is a *steady-state* algorithm, a class of MOEAs that only replaces one solution in the population during each full mating, mutation, and selection iteration of the algorithm. This is in contrast to *generational* algorithms, which replace the entire population in a single iteration. The ε-MOEA directly incorporates ε-dominance archiving into its search process by ensuring that mating and mutation always occur using an archived solution and a member of its search population.

In contrast, the ε-NSGAII [18] is a generational algorithm that exploits an ε-dominance archive. The ε-NSGAII also features adaptive population sizing and time continuation [64]. The term "time continuation" refers to the injection of diverse new solutions to reinvigorate search. Whereas many MOEAs used fixed population sizes throughout an entire run, the ε-NSGAII's adaptive population sizing and time continuation is produced using a series of smaller connected runs, where the population size is adjusted between runs. This population size adjustment improves search quality by maintaining a population size relative to the Pareto approximate set size. The ε-NSGAII's adaptive search seeks to exploit pre-conditioning to attain high quality Pareto approximation sets efficiently.

The OMOPSO [65] is the only particle swarm optimization (PSO) algorithm tested in this study and remains one of the most successful PSO algorithms to date. It is notable for being the first multiobjective PSO algorithm to include ε-dominance as a means to solve many-objective problems.

3.3. Indicator-Based Methods (IBEA)

Traditional MOEAs exploit nondomination sorting in their evolution of Pareto approximate solutions. As discussed in Section 1, this follows Goldberg's recommendation to exploit evolutionary optimization to minimize the number of solutions that dominate the population's performance in all objectives (i.e., seek nondominated solutions). Alternatively, indicator-based MOEAs replace nondomination sorting with alternative "metrics" of multiobjective performance. The most commonly employed metrics include hypervolume and ε-indicator to optimize the relative quality of evolving approximations to the Pareto front (see the discussion of these metrics in Section 4.2). In this study, we test one of the best known and most widely used of the indicator MOEAs, the Hypervolume Indicator-Based Evolutionary Algorithm (IBEA) [66]. The algorithm optimizes the "volume" of the objective space dominated by the solution generated with mating and mutation operators (see Fig. 2 and Eq. (4) in Section 4.2). It should be noted that because hypervolume is computationally demanding to compute for many-objective problems, IBEA's algorithmic computational demands can be significant, even for problems with expensive function evaluations.

3.4. Aggregate functions (MOEA/D)

Similar in intent to indicator-based methods, aggregate functions also convert many-objective problems into a single-objective

problem using an aggregation function. The MOEA/D [67] is a recently-introduced MOEA that uses aggregate functions, but attempts to avoid the pitfalls in prior aggregation approaches by simultaneously solving many single-objective Chebyshev decompositions of many-objective problems in a single run. Since its introduction, the MOEA/D has established itself as a benchmark for new MOEAs by winning the 2009 IEEE Congress on Evolutionary Computation (CEC 2009) competition [68].

3.5. Differential evolution (GDE3)

The GDE3 [69] is a multiobjective variant of the differential evolution (DE) algorithm [70]. The GDE3 (and DE in general) has the core distinguishing feature that its mutation operator uses the scaled "difference" between two population members' decision variable vectors to generated new candidate solutions. It is expected early in the evolutionary process that these differences are large and gradually reduce as the population converges. A unique mathematical consequence and benefit of the operator is that it is termed rotationally invariant. Any variation operator that is "rotationally invariant" does not assume explicit search directions when it creates new solutions. The use of DE also means that GDE3 does not require decisions to be separable and independent (i.e., they can have conditional dependence). Alternatively, most modern MOEAs use simulated binary crossover (SBX) and polynomial mutation (PM) as shown in Table 1. The SBX operator assumes problems have independent decisions that can be optimized using only vertical or horizontal translations of their decision variables. This is a major assumption in MOEAs that has largely been ignored in the MOEA literature. The GDE3 was one of the top rated MOEAs in the CEC 2009 competition [68]. Readers should note that MOEA/D also uses the DE search operator.

3.6. Adaptive operator selection (AMALGAM, Borg)

Given the variety of fitness landscapes and the complexity of search population dynamics, Vrugt et al. [71,72] proposed adapting the evolutionary operators used during multiobjective search based on their success in guiding search. In this study, we use the multiobjective AMALGAM MOEA [71], which as its name implies, exploits multi-algorithm search that combines the NSGAII, PSO, DE, and adaptive metropolis (AM). AMALGAM represents a major contribution due to its recognition for the need to combine multiple search operators when addressing a broad suite of challenging problems. AMALGAM has had an increasing presence in the water resources literature and this study provides the first rigorous assessment of its capabilities relative to other state-of-the-art MOEAs.

The Borg MOEA [16,44] combines adaptive operator selection with ε-dominance, adaptive population sizing and time continuation. The resulting algorithm is designed for handling many-objective, multimodal problems from a wide assortment of problem domains. The Borg MOEA uses the simulated binary crossover (SBX), differential evolution (DE), parent-centric recombination (PCX), unimodal normal distribution crossover (UNDX), simplex crossover (SPX), polynomial mutation (PM) and uniform mutation (UM) operators. It should be noted that several of Borg's operators are rotationally invariant (see the discussion of this topic in Section 3.5). Since each operator inherently favors certain fitness landscapes, Hadka and Reed [44] demonstrated that the Borg MOEA is able to quickly select the most appropriate operators for a given problem. Another unique feature of Borg is ε-progress. This feature tracks changes in the ε-dominance archive, allowing the algorithm to quickly detect search stagnation. Upon detecting stagnation (i.e., no improvements found during a fixed time period), Borg performs a restart to inject new diverse solutions into the search population. A similar restart strategy is used in the ε-NSGAII [18]. This study builds on a comprehensive diagnostic assessment of MOEAs for the most challenging test functions in the evolutionary computation literature [16] in which Borg was the top overall algorithm.

Table 1
Detailed summary of the latin hypercube statistical sampling of the MOEAs' operators and their associated parameter ranges for each of the applications.

	Parameter/problem	Range	Algorithms
Crossover	SBX Rate	0.0–1.0	Borg, AMALGAM, IBEA, ε-MOEA, ε-NSGAII,SPEA2, NSGAII
	SBX Distribution Index	0.0–500.0	Borg, AMALGAM, IBEA, ε-MOEA, ε-NSGAII, SPEA2, NSGAII
	DE Crossover Rate	0.0–1.0	Borg, AMALGAM, GDE3, MOEA/D, NSGAII
	DE Step Size	0.0–1.0	Borg, AMALGAM, GDE3, MOEA/D
	PCX Parents	2.0–10.0	Borg
	PCX Offspring	1.0–10.0	Borg
	PCX Eta	0.0–1.0	Borg
	PCX Zeta	0.0–1.0	Borg
	UNDX Parents	2.0–10.0	Borg
	UNDX Offspring	1.0–10.0	Borg
	UNDX Eta	0.0–1.0	Borg
	UNDX Zeta	0.0–1.0	Borg
	SPX Parents	2.0–10.0	Borg
	SPX Offspring	1.0–10.0	Borg
	SPX Epsilon	0.0–1.0	Borg
	Metro Jump Rate	0.5–5.0	AMALGAM
	PSOC1	0.0–2.0	AMALGAM
	PSOC2	0.0–2.0	AMALGAM
	PSO Chi	0.0–1.0	AMALGAM
Mutation	PM Rate	0.0–1.0	Borg, AMALGAM, MOEA/D, IBEA, ε-MOEA, ε-NSGAII, SPEA2, NSGAII
	PM Distribution Index	0.0–500.0	Borg, AMALGAM, MOEA/D, IBEA, ε-MOEA, ε-NSGAII, SPEA2, NSGAII
	UM Rate	0.0–1.0	Borg
	Perturbation Index	0.0–1.0	OMOPSO
Selection	Neighborhood Size	0.0–0.2	MOEA/D
	Delta	0.0–1.0	MOEA/D
	Eta	0.0–0.02	MOEA/D
	Archive Size	10–1000	IBEA, OMOPSO, SPEA2
	Injection Rate	0.1–1.0	Borg, ε-NSGAII
PopSize	DTLZ2	10–1000	All Algorithms
	HBV	10–1000	All Algorithms
	LTM	10–1000	All Algorithms
	LRGV	10–1000	All Algorithms
NFE	DTLZ2	$10^4 - 10^6$	All Algorithms
	HBV	$10^4 - 2 \times 10^2$	All Algorithms
	LTM	$10^4 - 10^5$	All Algorithms
	LRGV	$10^4 - 10^5$	All Algorithms
LHS	DTLZ2	1000	All Algorithms
	HBV	500	All Algorithms
	LTM	500	All Algorithms
	LRGV	100	All Algorithms

4. Diagnostic framework

4.1. Experimental overview

Fig. 1 adapted from Hadka and Reed [16] provides a schematic overview of the diagnostic assessment framework used in this

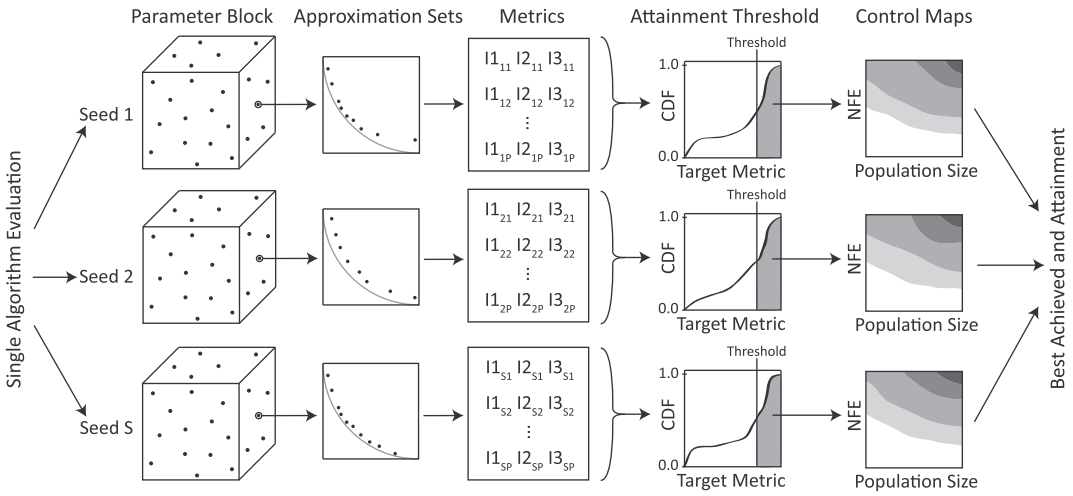


Fig. 1. Illustrative overview of the diagnostic assessment performed for each MOEA (adapted from Hadka and Reed [16]).

study to statistically compare the ten algorithms discussed in Section 3. The figure illustrates the approach we followed for each algorithm. MOEAs are stochastic search tools that range in their use and parameterization of operators (i.e., mating, mutation, selection, archiving, etc). Our diagnostic assessment avoids any specific assumption on "optimal" or "highly-tuned" parameters for each algorithm. Instead, as depicted in the parameter blocks shown in Fig. 1, Latin hypercube sampling (LHS) was used to sample the full feasible parameter space for each algorithm. Each point in the parameter blocks represents a full specification of the MOEA's parameters that are run to attain its corresponding approximation set for a given problem. Since algorithm performance can be affected by random effects in initial populations and operators, the runs of the LHS parameter block were replicated for 50 random number generator seeds. For each approximation set, several computed metrics measure its convergence, consistency, and diversity (see Section 4.2). For the water resources test problems, the true "global" Pareto fronts are unknown. Consequently, we generated our best approximations of the problems' Pareto fronts across all runs of all of the algorithms. These best known approximations served as the reference sets in our metrics calculations.

Fig. 1 also shows that attainment thresholds (i.e., goal metric values) were used to quantify the probabilities of successful search across all runs. This is the first study to statistically characterize the "controllability" of MOEAs for water resources applications. The term controllability refers to the fact that parameterization choices can have significant impacts on the success or failure for an algorithm. Goldberg [64] clarifies that when an algorithm is well suited to a problem, almost any parameter combination will yield acceptable performance. He referred to this as a large "sweet spot" in the parameter space of an algorithm. Alternatively, for highly challenging problems, selecting a successful parameterization may be extremely difficult if not impossible (i.e., the MOEA is not controllable). To convey this concept, Fig. 1 illustrates that we utilize control maps developed from our LHS of parameterizations to show a sub-space projection plot of performance for population size versus the number of function evaluations (NFE) used in search. Overall population size has a dominant impact on MOEA performance and NFE is critical for understanding their efficiency given computational constraints. Although we could have shown other parameter combinations, these have the broadest impact and appeal for the practical implementation of MOEAs. The control maps depict "sweet spots" based on how each sampled parameterization performs across its 50 random seed trials.

Table 1 provides a detailed summary of the operators, parameter ranges, and LHS sampling for our analysis. This study represents the largest and most rigorous assessment of MOEAs ever completed for the water resources area. It is also the first study to extend the comprehensive MOEA diagnostic assessment framework by Hadka and Reed [16] to real-world applications. Overall, this effort encompasses over 100,000 MOEA runs requiring trillions of design evaluations to provide a comprehensive assessment of the state-of-the-field for water resources applications.

4.2. Metrics for convergence, consistency, and diversity

Convergence and diversity are critical measures of MOEA performance. In single objective optimization, it is straight forward to monitor the progress of search by simply monitoring the dynamic improvement of the objective. In multiobjective optimization, this issue is far more complex, as has been carefully discussed in the work by Knowles and Corne [73] and Zitzler et al. [74]. When evaluating the search quality and progress of MOEAs, it is important to consider the proximity of an approximation set to the Pareto optimal front or its best known approximation (i.e., convergence to the reference set) as well as its representation of the full extent of tradeoffs (i.e., diversity – see Fig. 2a). The evolutionary computation literature clearly conveys that multiple measures are necessary to understand these properties. Likewise, recent comparative studies in the water resources area also demonstrate the need for multiple measures [17–19,75].

Although we utilized a very large suite of performance metrics, our results and analysis will focus on three performance measures: (1) generational distance [76], (2) additive ε-indicator [74], and (3) hypervolume [74]. Each of these measures provides unique insights into the performance of MOEAs.

The generational distance measure [76] is the easiest of the metrics to satisfy, requiring only a single solution to be close to the reference set (a pure measure of convergence as illustrated in Fig. 2b). Overall it measures the average Euclidean distance of points in an approximation set to their nearest corresponding objective vectors in the reference set. The generational distance, I_G, is measured between the local Pareto approximate set, $\mathcal{PF}$, and the reference (or best known) Pareto approximate set, $\mathcal{PF}'$, and is defined by Eq. (2)

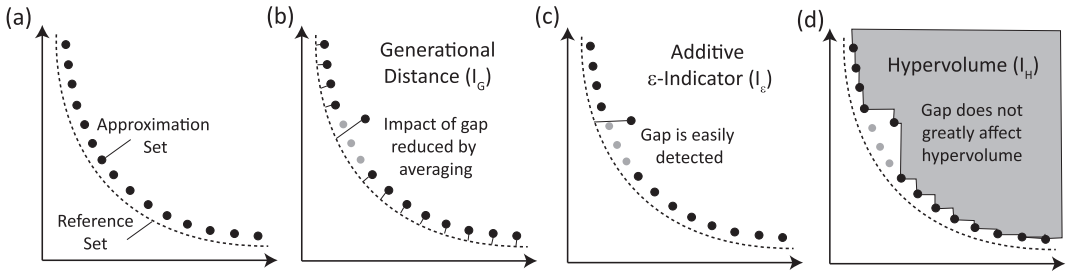


Fig. 2. Illustration of how different metrics capture convergence, diversity, and consistency (i.e., absence of gaps). (a) A good approximation to the reference set, is indicated by the dashed line. (b) Generational distance averages the distance between the approximation set and reference set, requiring only one point near the reference set to attain ideal performance. The gap in the approximation set (indicated by the missing light gray points) has no impact because other solutions are near the reference set. (c) The ε-indicator metric easily identifies the gap (yielding a metric value 2–3 times worse) because it focuses on the worst case distance required to translate a point to dominate its nearest neighbor in the reference set. (d) However, the change in hypervolume due to a gap is small relative to the entire hypervolume. Hypervolume rigorously captures both convergence and diversity making it the most difficult metric to satisfy. Figure adapted from Hadka and Reed [16].

$$I_G(\mathcal{PF},\mathcal{PF}') = \frac{\sqrt{\sum_{\mathbf{x}\in\mathcal{PF}} d_{\mathbf{x}}^2}}{P} \quad \text{where} \quad d_{\mathbf{x}} = \min_{\mathbf{y}\in\mathcal{PF}'} \sqrt{\sum_{i=1}^{M} [f_i(\mathbf{x}) - f_i(\mathbf{y})]^2}, \quad (2)$$

where $d_{\mathbf{x}}$ is the minimum Euclidean distance between each of the objective vectors in the local Pareto approximate set $\mathcal{PF}$, and each of the objective vectors in the reference Pareto approximate set $\mathcal{PF}'$.

Alternatively, the additive ε-indicator [74] measures the worst case distance (versus the average distance) required to translate an approximation set solution to dominate its nearest neighbor in the reference set. The additive ε-indicator, I_ε, between the local Pareto approximate set, $\mathcal{PF}$, and the reference (or best known) Pareto approximate set, $\mathcal{PF}'$ is defined by Eq. (3)

$$I_\varepsilon(\mathcal{PF},\mathcal{PF}') = \max_{\mathbf{y}\in\mathcal{PF}'} \min_{\mathbf{x}\in\mathcal{PF}} \max_{1\leqslant i\leqslant M} (f_i(\mathbf{x}) - f_i(\mathbf{y})), \quad (3)$$

where objective values $f_i(\mathbf{x})$ and $f_i(\mathbf{y})$ from each of the Pareto approximate sets $\mathcal{PF}$ and $\mathcal{PF}'$ respectively are compared across all M objectives. As explained in Hadka and Reed [16], the additive ε-indicator is very sensitive to gaps in tradeoffs and can be viewed as a measure of an approximation set's "consistency" with the reference set (i.e., all portions of the tradeoff are present as illustrated in Fig. 2c). The additive ε-indicator is more influenced by gaps and the diversity of the approximation set than its convergence. Additionally, the measure captures diversity because of its focus on the worst case distance. If a Pareto approximate set, $\mathcal{PF}$, has gaps, then solutions from other regions must be translated much further distances, dramatically increasing the additive ε-indicator metric value (see Fig. 2c).

The hypervolume is the most challenging of the three measures to satisfy. As illustrated in Fig. 2d, hypervolume measures the volume of objective space dominated by an approximation set. The hypervolume indicator I_H is calculated as the difference in hypervolume between the reference (or best known) Pareto approximate set $\mathcal{PF}'$, and the local Pareto approximate set $\mathcal{PF}$ as defined in Eq. (4).

$$I_H(\mathcal{PF},\mathcal{PF}') = \frac{\int_V \alpha_{\mathcal{PF}}(\mathbf{x})dx}{\int_V \alpha_{\mathcal{PF}'}(\mathbf{y})dy} \quad \text{where}$$
$$\alpha_{\mathcal{PF}}(\mathbf{x}) = \begin{cases} 1 & \text{if } \exists\ \mathbf{z}\in\mathcal{PF} \text{ such that } \mathbf{z}\preceq\mathbf{x} \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

The hypervolume calculation is performed across the normalized objective function values, hence the volume integral denoted by $\int_V$, is calculated between the reference points 0 to 1. The *attainment function*, $\alpha(\mathbf{x})$ defines the portion of the objective space that is dominated by the objective vectors in either the local or reference Pareto approximate sets. Knowles and Corne [73] highlight that the hypervolume indicator respects the dominance relationships between any two solution sets being compared (i.e., good performance means a solution set is both converged and diverse).

5. Test problems

5.1. DTLZ2

The overwhelming majority of MOEA studies to date have explored problems with two or three objectives [2]. However, with many real-world applications involving four or more objectives (see the following reviews and applications [2,50,30,39,29,41,12,1,15]), it is necessary to understand the limits of MOEA scalability. To this end, we explore the scalability of ten MOEAs on a relatively simple test problem, DTLZ2, with the aim of identifying the upper bound number of objectives an MOEA supports. The DTLZ2 problem was introduced in the scalable test problem suite by Deb et al. [77]. The problem's formulation is shown in Eq. (5) and a three-objective instance of the problem is visualized in Fig. 3.

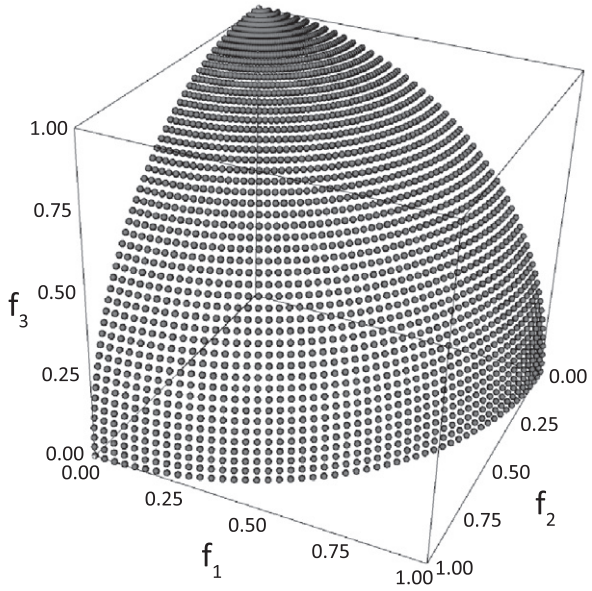


Fig. 3. Three objective instance of the DTLZ2 test function used in our analysis of the MOEAs' scalability with increasing objective counts.

$$
\begin{aligned}
&\text{Minimize}: \\
&f_1(\mathbf{x}) = (1+g(\mathbf{x}_M))cos(x_1\pi/2)cos(x_2\pi/2)\ldots \\
&\qquad cos(x_{M-2}\pi/2)cos(x_{M-1}\pi/2) \\
&f_2(\mathbf{x}) = (1+g(\mathbf{x}_M))cos(x_1\pi/2)cos(x_2\pi/2)\ldots \\
&\qquad cos(x_{M-2}\pi/2)sin(x_{M-1}\pi/2) \\
&f_3(\mathbf{x}) = (1+g(\mathbf{x}_M))cos(x_1\pi/2)cos(x_2\pi/2)\ldots \\
&\qquad sin(x_{M-2}\pi/2) \\
&\vdots \qquad\qquad \vdots \\
&f_{M-1}(\mathbf{x}) = (1+g(\mathbf{x}_M))cos(x_1\pi/2)sin(x_2\pi/2), \\
&f_M(\mathbf{x}) = (1+g(\mathbf{x}_M))sin(x_1\pi/2) \\
&\text{Subject to} \quad 0 \leqslant x_k \leqslant 1 \forall k = 1,2,\ldots,L, \\
&\text{where } g(\mathbf{x}_M) = \sum_{x_k \in \mathbf{x}_M} (x_k - 0.5)^2.
\end{aligned}
\tag{5}
$$

In Eq. (5), there are M objective functions, and the number of decision variables utilized is dependent on M according to $L = M + 9$ [78]. The DTLZ2 problem exhibits several key characteristics for scalability analysis: the decision variables are separable (i.e., they can be optimized independently), the search space is continuous and unimodal, and the problem is not deceptive (i.e., improvements in decisions from anywhere in the problem's search space lead to a single global Pareto front). These characteristics are considered easy for any modern MOEA. This implies that any performance degradation can be attributed to the scalability limitations of the MOEA and not the inherent difficulty of the problem [33,34].

In this study, the DTLZ2 test problem was analyzed for between 2 and 16 objectives. Our results for the DTLZ2 test case will focus on the generational distance and additive ε-indicator metrics solely. The hypervolume metric's computational demands are too severe for high objective counts (i.e., more than eight objectives).

5.2. Rainfall–runoff model calibration

The HBV model is a lumped conceptual rainfall–runoff model that has been applied to flow prediction problems worldwide [79]. The model consists of three primary routines: (i) snow accumulation and melt, represented by a variant of the degree-day snow model, (ii) soil moisture accounting, in which a probability distribution is used to represent the spatial variability of storage elements and (iii) a response routine, in which the linear outflow from two sub-basins is transformed by a routing parameter. In sum, the model contains 14 real-valued decision variables that require calibration.

The MOEAs in this study were used to calibrate the HBV model for the Williams River, West Virginia, United States (USGS Gage 03186500), which was selected based on a screening analysis judging the difficulty of calibration problems across 392 watersheds within the MOPEX dataset developed by Duan et al. [80]. The Williams River is located in the Appalachian Mountains and thus contains a challenging combination of steep slopes and snow-driven runoff. The model itself provides an additional degree of difficulty with its high number of parameters and strong parameter interactions.

Calibration was performed using precipitation and streamflow data from the MOPEX dataset over the period 1962–1972 with a one-year warmup period. Four objective functions were used: the Nash–Sutcliffe Efficiency (NSE), which measures the model fit primarily during high-flow periods; the Box-Cox transformed root mean squared error (TRMSE), which accounts for low-flow periods; the runoff coefficient error (ROCE), which accounts for the long-term water balance; and the slope of the flow duration curve error (SFDCE), which measures the ability of the model to match the long-term variability of flows. The calculations for these functions are described in detail in van Werkhoven et al. [81] and Wagener et al. [82].

Several recent studies have shown that in general, rainfall–runoff calibration problems can cause MOEAs to fail to attain high quality Pareto approximate sets in a limited number of function evaluations [19,15]. Tang et al. [19] explicitly illustrated "false fronts" (i.e., multi-modality) whose model parameterizations are in different regions of the search space from the best known Pareto approximate set (i.e., deceptive to search). More generally, Hadka and Reed [16] show that multi-modality is a severe challenge for most MOEAs. Mathematically, the HBV calibration test case represents a nonlinear and multimodal problem (i.e., the objective space contains false optima). An overview of the problem formulation and its properties is shown in Table 2.

5.3. Groundwater monitoring design

The LTM case study [18,83] represents a complex groundwater monitoring application in which 58 sampling ports are available to monitor a perchloroethylene (PCE) contamination plume originating from an underground storage tank. The study site is hypothetically drawn from a plume located at the Lawrence Livermore National Lab (LLNL) within a highly heterogeneous alluvial aquifer, the hydrogeology of which has been characterized extensively. The PCE plume represents a synthetic test case generated using a 50 million node flow and transport simulation at the LLNL site [84]. The objective of this case study is to optimally choose a subset of the 58 sampling locations that meet four performance objectives simultaneously, all of which are minimized.

The four performance objectives include minimizing the cost of sampling the system (Cost), minimizing the PCE concentration estimation error (Error), minimizing the PCE concentration estimation uncertainty (Uncertainty), and minimizing the error in estimating the mass of PCE present at the site (Mass). The Error and Mass objectives are calculated using Quantile Kriging (QK), an extension of Ordinary Kriging [85], whereby a "maximum data" baseline is established by generating Kriged maps of PCE concentration and mass using all available sampling locations. The error for these objectives is then calculated as an error (aggregated across the Kriged locations) relative to this baseline for each

Table 2
Summary of the formulations and properties for the water resources problem suite.

Problem	Decisions (# of variables)	Objectives	ε Search Precision	Constraints	Properties
HBV	Model Parameters (14)	Min. NSE, Min. TRMSE, Min. ROCE, Min. SFDCE	0.01, 0.025, 1.0, 1.0	N/A	Real-valued decisions, Multi-modal, Deceptive, Non-linear
LTM	Sampling Points (58)	Min. Cost, Min. Error, Min. Uncertainty, Min. Mass	1.0, 2.0, 10.0, 0.33	No unestimated grid points	Discrete decisions Large decision space, Inter-dependent decisions, Constrained
LRGV	Rights (1) Options (3) Supply/Demand Thresholds (4)	Min. Cost, Max. Reliability, Min. Surplus, Min. Dropped Transfers, Min. Number of Leases	0.0009, 0.002, 0.03, 0.004, 0.004	Reliability > 0.98 Critical Reliability = 1.0 Cost Variability < 1.1	Real-valued and integer decisions, Stochastic objectives, Constrained

candidate sampling plan that utilizes a subset of the sampling locations. The Uncertainty objective represents the summation of the estimation variances that directly result from the Kriging process at each of the Kriged locations throughout the domain. The four-objective problem is constrained such that sampling plans must provide a sufficient spatial coverage of the full extent of the plume so that interpolation is possible at every grid point. The LTM problem represents a constrained, discrete combinatorial search problem with 2^{58} possible designs. All of the MOEAs uniformly implemented a penalty-based constraint formulation for the LTM test case, which has been studied extensively in the past (please refer to the following studies [28,18,26,83]).

Table 2 provides a synopsis of the LTM formulation and its properties. The LTM problem has been shown to be severely challenging for traditional MOEAs [83] due to the strong interdependencies in the monitoring decisions (e.g., it is important to sample the boundaries and the source area of a plume to attain accurate mass estimates). In fact, this study builds on Kollat et al. [83] to confirm that traditional MOEA search operators are not capable of rigorously converging to a diverse representation of the LTM test case's four-objective reference set. This case study illustrates the "epistasis" failure mode where traditional MOEA recombination and mutation operators (SBX, PCX, DE, UNDX, SPX, PSO, PM, UM) cannot capture complex statistical relationships between decision variables and consequently will not lead to high attainment. Our hypothesis in including the binary coded LTM test case is that the best overall performance of every MOEA tested in this study will fall short in attaining high quality values for the hypervolume and ε-indicator metrics. To date, only an advanced class of Bayesian optimization MOEA as introduced in Kollat et al. [83] has been able to reliably converge to highly diverse Pareto approximate sets for the LTM test case. Note that we have not included the more advanced class of MOEA in this study as they are not generally applicable to the integer and real-valued decisions in the HBV and LRGV test cases.

5.4. Water supply portfolio planning

The Lower Rio Grande Valley (LRGV) case study is a risk-based water supply portfolio management problem that seeks to help a single city identify the most efficient combination of market-based and traditional reservoir sources for its water supply. Since the problem is highly-constrained and evaluated with respect to a stochastic simulation, it represents a challenging benchmark for the MOEAs. The city's water supply portfolio consists of three supply instruments: non market-based permanent rights to reservoir inflows, spot leases acquired at any month in the year with a variable price, and an adaptive options contract which reduces lease price volatility by guaranteeing a fixed price for water acquisitions made at a specified point later in the year. The portfolio has eight real-valued decision variables which control rights (one variable), options (three variables), and thresholds used to determine market acquisitions based on available data on supply and demand (four variables). A Monte Carlo simulation performs random monthly draws that simulate 10 years of water supply decisions. The calculations track reservoir mass balance, municipal demand, lease pricing distribution, and portfolio performance for each of the 120 simulation months and replicate this process many times to calculate expected performance for the objectives. The objectives in this study follow the most restrictive Case D formulation in Kasprzyk et al. [39], with ε values set following the noise-adjusted ε procedure in Kasprzyk et al. [42]. Table 2 provides and overview of the LRGV formulation. Each function call comprised 1000 Monte Carlo samples of the simulation model. Additional information on the simulation model is provided in [86,87].

The LRGV is the only problem tested in this study to use explicitly defined constraints (in contrast to the penalty function of the LTM test case). All of the tested MOEAs except MOEA/D use restricted tournament selection [88]. In restricted tournament selection, the selection operator always favors individuals which do not violate constraints. If both solutions do not violate constraints, Pareto dominance is the criteria used during selection. In the case where one or both individuals violate constraints, the individual with the least constraint violation is preferred. MOEA/D uses a similar strategy, which was tailored to MOEA/D's unique architecture. Since MOEA/D uses neighborhood-based selection, which does not explicitly use a selection operator, constraints are applied when determining if offspring survive and replace existing members of the population. Identical to restricted tournament selection, replacement favors individuals which do not violate constraints.

The LRGV represents a highly challenging search problem for several reasons. Its stochastic objectives and constraints are "noisy", which can lead to deterioration failures for some MOEAs (see discussion of deterioration in Section 2.1). Kasprzyk et al. [42] provides a more detailed discussion of the potentially negative impacts of the LRGV objectives' uncertainties on MOEA search. Singh et al. [57] also explicitly illustrate how MOEA convergence can be adversely impacted with uncertain objectives. The LRGV problem also has decisions that combine large integers and real-valued variables to yield a highly irregular Pareto front geometry with two very different regions of solutions in terms of their objective performance and underlying design characteristics. This irregular geometry makes it very difficult for MOEAs unless they strongly maintain the diversity of their search populations. Lastly, to assess the difficulty of LRGV's constraints, we executed a random baseline where we used a LHS sample of 500,000 potential designs and determined that only 0.06% of the randomly drawn LRGV solutions were feasible. Moreover, the feasible solutions discovered by the LHS were all of poor quality relative to our MOEA generated Pareto approximate reference set for the problem. Note that all of the metrics described in Section 4.2 were computed relative to the mean objective values of the reference set solutions, which were confirmed with an independent draw of 100,000 Monte Carlo LRGV simulations for every solution.

6. Results and discussion

6.1. DTLZ2 objective scaling test

Fig. 4 provides results for the scalability of the MOEAs on instances of the DLTZ2 test problem with increasing objective counts. In the figure, the algorithms are listed in succession on the horizontal axis and the objective count for DTLZ2 is shown on the vertical axis. Black shading represents ideal performance and white shading represents failure. The term scalability in this context refers to the maximum objective count where a given algorithm is able to still attain some level of performance with respect to the generational distance and ε-indicator measures. Note that we did not use the hypervolume measure for these results because with more than nine objectives, this metric becomes too computationally intensive given the large number of runs analyzed. Recall that the generational distance metric is the easiest to satisfy and only requires a single point be near the known reference Pareto front. The ε-indicator metric is more challenging to satisfy and is highly sensitive to gaps in the approximation set (see Fig. 2). Fig. 4a shows the best performance attained for each algorithm across all of its LHS parameterizations and random trials.

In Fig. 4a, the generational distance results highlight a separation in the MOEAs performance beginning at seven objectives. To this point, all of the MOEAs' best runs were at least able to attain

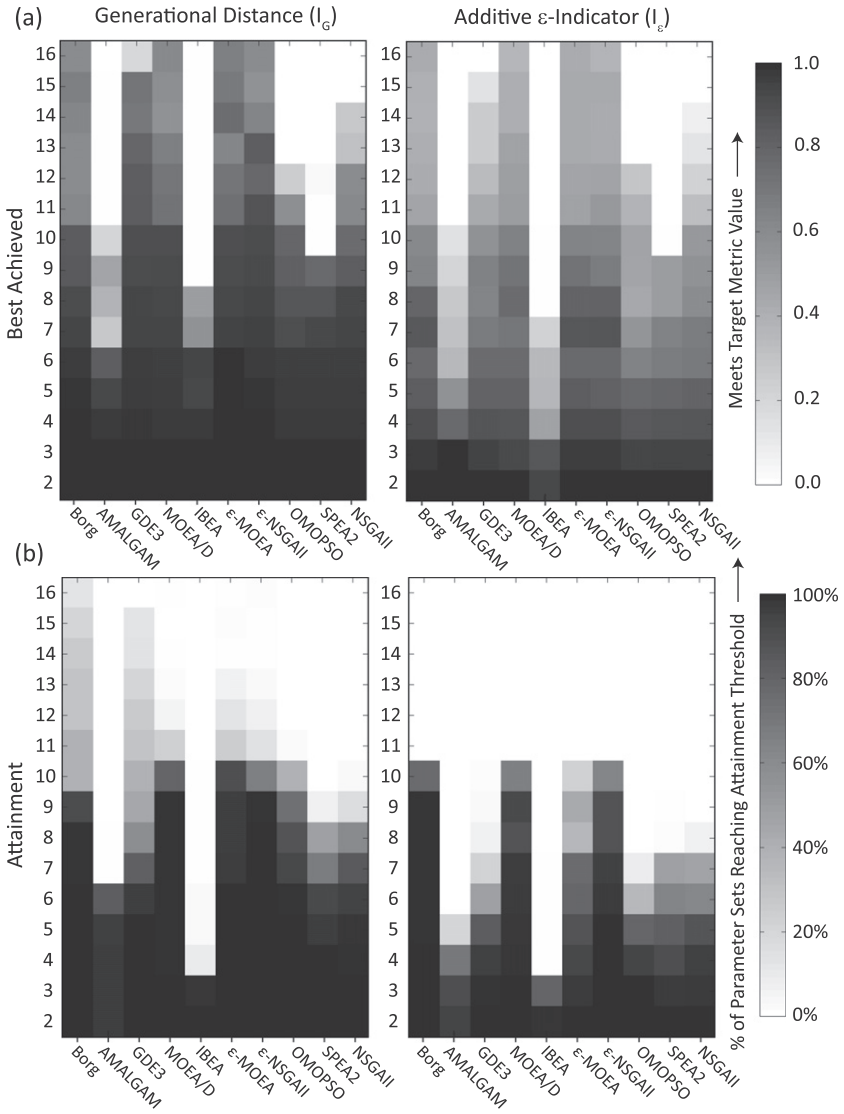


Fig. 4. Scalability of the algorithms evaluated in terms of (a) their best performance across all runs and (b) the probability of attaining 50-percent of the ideal metric values.

one or more points near their true Pareto fronts. IBEA, AMALGAM, and SPEA2 fail beyond 10 objectives. The best overall performers with respect to generational distance are Borg, GDE3, MOEA/D, ε-MOEA, and ε-NSGAII. When accounting for the ε-indicator results in Fig. 4a, the best single runs of Borg, MOEA/D, ε-MOEA, and ε-NSGAII were still able to attain some level of success even at objective counts as high as 16.

Although a single run for an algorithm may be successful, a critical concern is the probability of good performance. Fig. 4b captures attainment success probabilities for 50-percent of the reference metric value. This is a far more challenging measure of performance. If an algorithm exhibits very good performance with respect to its best single run, but only a small percentage of runs attain this good performance, it would be very difficult for users to implement effectively. In this situation, finding an effective parameterization is equivalent to a "needle-in-a-haystack" problem. Fig. 4b shows that this phenomenon occurs for all of the algorithms for instances of DTLZ2 that have more than 10 objectives. In the context of generational distance, Borg and GDE3 still maintained some probability of attaining the target performance threshold for instances of the DTLZ2 problem with up to 16 objectives. Although GDE3 is able to maintain some probability of attainment for the generational distance metric, the algorithm largely fails to maintain high probabilities of attainment for the ε-indicator results at objective counts exceeding six objectives. This means that although the GDE3 is finding some solutions across its sampled parameterizations, they are not consistent with the DTLZ2 reference front (i.e., they have significant gaps).

As the top performers for the ε-indicator attainment results, Borg, MOEA/D, and ε-NSGAII were still able to maintain a high probability of success at 10 objectives. It is worth noting that this 10 objective bound on attainment matches the theoretical expectations of Teytaud [31,37]. Teytaud's work highlights that as objective count increases linearly, the percentage of a problem's space that is non-dominated grows exponentially. Beyond 10-objectives the exponential growth of the non-dominated region makes it highly challenging for MOEAs to maintain reliable and controllable search. Fig. 4b indicates that several of the algorithms can maintain convergence (i.e., high generational distance attainment) beyond 10 objectives but not diversity (i.e., poor ε-indicator attainment). Both Borg and ε-NSGAII benefit from their adaptive population sizing operators when attaining effective and reliable many-objective search. MOEA/D's neighborhood-based Chebyshev decomposition is quite effective in facilitating search. However, it is worth noting that MOEA/D is highly sensitive to its population size for challenging problems and its neighborhood-based search causes

the algorithm's computational demands to grow severely with population size. In some instances, MOEA/D may not be tractable if a large population is required [16]. High attainment is a key property for MOEAs in water resources applications, as it implies that a high percentage of an algorithm's parameter space yields effective search (i.e., convergence of a diverse set of solutions towards the Pareto front).

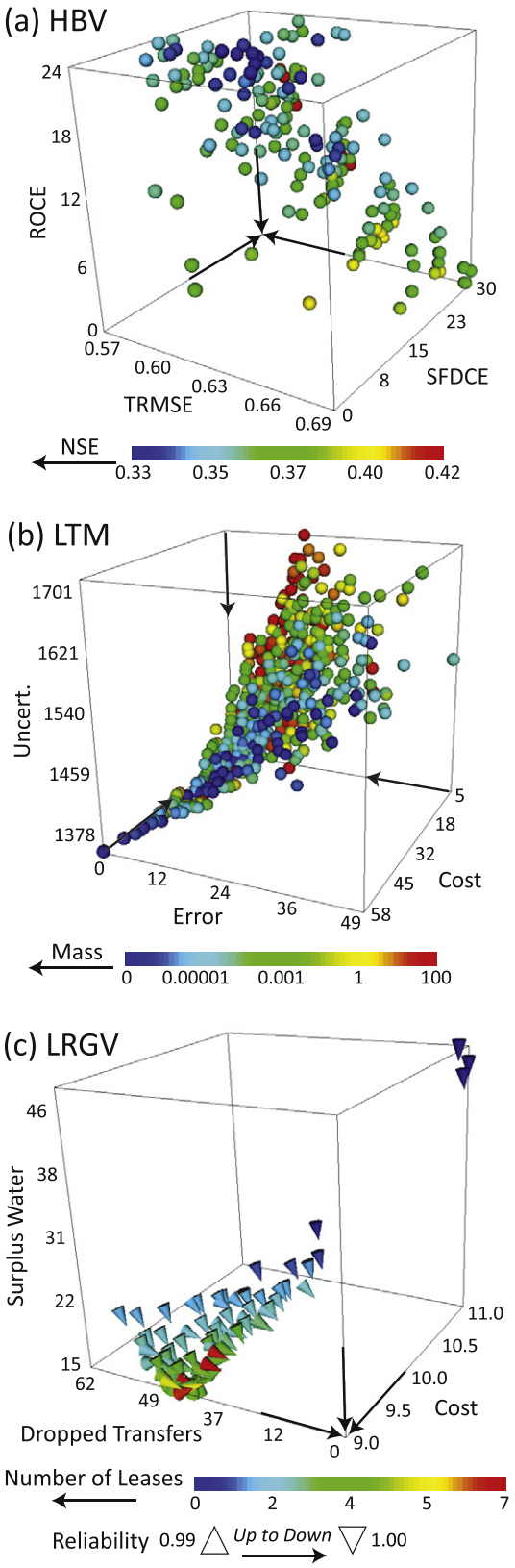


Fig. 5. Illustration of best known reference Pareto fronts attained across all runs of all algorithms for (a) the HBV calibration test case, (b) the LTM design problem, and (c) the LRGV water management test case. The black arrows indicate direction of increasing preference.

6.2. Water resources applications

Unlike DTLZ2, where the analytical Pareto front is known, the HBV calibration, LTM, and LRGV water management applications have unknown Pareto fronts. Fig. 5a–c provide visualizations of the best known reference sets attained for these problems across all runs of all of the MOEAs tested. The geometries of the tradeoffs vary significantly across the applications, as would be expected given their different defining properties. In each of the plots, black arrows have been added to guide the reader in understanding the directions of increasing preference. For the calibration and monitoring problems' reference sets illustrated in Fig. 5a and b, respectively, their four objectives are captured with three primary axes and a color legend. The LRGV water management application is a five-objective formulation where in addition to the three primary plot axes and color, the cone's orientation illustrates performance with respect to reliability (down is highly reliable; up is lower reliability in Fig. 5c).

As summarized in Table 2, this suite of water applications captures a broad range of challenging mathematical properties. The HBV calibration problem is multi-modal, deceptive, and nonlinear. The LTM problem is constrained, discrete, and possesses a very large combinatorial decision space (i.e., 2^{58} potential designs). The LRGV risk management application is the most computationally demanding due to its stochastic objectives and severe constraints. In Fig. 5c, as noted in prior results [39,42], the LRGV has mixed real and integer decisions which cause the discontinuity between high surplus water solutions (shown in dark blue cones in the top right corner) and low surplus solutions (shown as cyan, green, and red cones in the lower left corner). These two distinct regions and the irregular geometry of the LRGV reference set make it particularly difficult to capture in terms of convergence and diversity.

Fig. 6 provides a measure of the percent of the reference set solutions that were captured by each MOEA within the ε-precisions specified in Table 2 across the three water resources applications. For a solution to be counted it must fall within the same ε-box as a given reference set solution. The percent of the reference sets captured by each algorithm for the HBV, LTM, and LRGV test cases

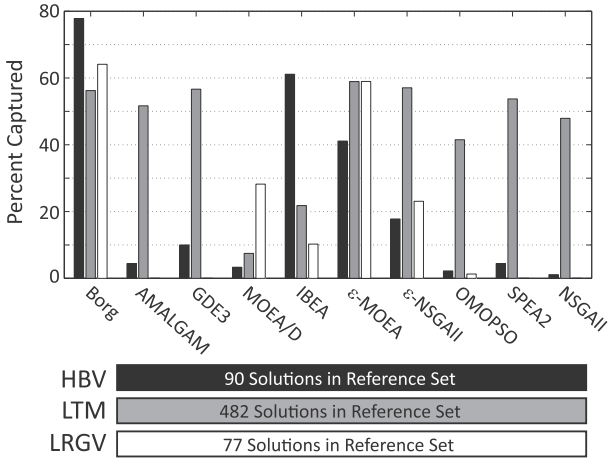


Fig. 6. Percent of the reference Pareto fronts captured (within the search ε's specified in Table 2) by each of the ten MOEAs for the HBV and LRGV test cases.

are shown in black, gray, and white, respectively. The percent captured by a given MOEA are computed across all of its runs for all of the sampled parameterizations summarized in Table 1 (i.e., each algorithm's best known Pareto approximate set). For the HBV calibration, Borg and IBEA captured more of the reference set shown in Fig. 5a than the remaining algorithms. ε-MOEA and ε-NSGAII captured between 18 to 42-percent of the HBV reference set. NSGAII, OMOPSO, SPEA2, and AMALGAM all struggled to capture a significant percent of the HBV reference set. In the LTM application, several of the algorithms were able to capture between 50-percent to 60-percent of the reference set solutions illustrated in Fig. 5b. IBEA and MOEA/D were the worst performing algorithms in terms of capturing the LTM reference set. The LRGV water management application was the hardest of the applications in terms of capturing the reference set in Fig. 5c. The LRGV reference set was largely contributed by Borg and ε-MOEA. AMALGAM, GDE3, SPEA2, and NSGAII failed to capture any of the LRGV test case's reference set. Overall, only the Borg algorithm consistently captured the highest percent of the reference sets across all of the water resources applications. Borg benefits from its adaptive search operators and its use of ε-progress to maintain effective search across different problems.

Readers should note that while capturing the reference set is important, the results in Fig. 6 must be evaluated in the broader context of probabilistic search performance to avoid false conclusions. Fig. 7 shows both the best overall performance and probabilistic attainments for all of the MOEAs across the three water resources applications. In the plot, each row represents a succession of results for a single application evaluated with respect to the generational distance, ε-indicator, and hypervolume metrics. Each column of the plot corresponds to the results for a single metric across the three different water applications. In each of the nine panels, the best single run performance is plotted as a black circle that indicates the percentage of the best possible metric value captured by that run. These results show the expected performance of the very best parameterization of each algorithm. The "best" performance assessment is particularly useful for identifying algorithms that fully fail to attain high quality approximations to the best known reference set. In addition to the MOEAs' best performance, Fig. 7 also shows the probability that each algorithm will attain a given percentage of the best possible metric value. This attainment probability is defined by the gray shading in Fig. 7's legend. Ideal performing MOEAs should have black circles and black shaded bars at 100% of the best metric value (e.g., generational

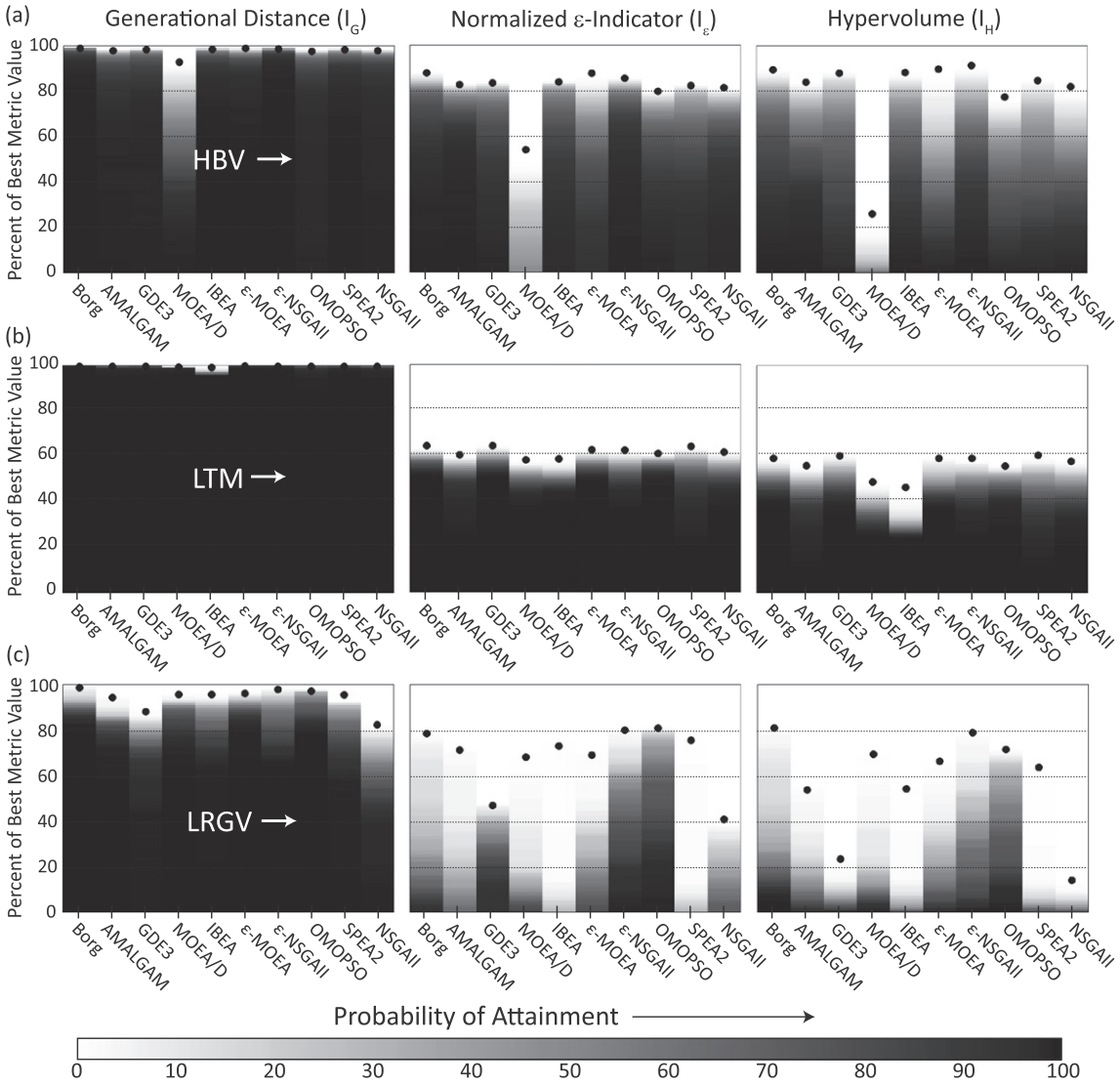


Fig. 7. Summary of the best overall performance attained by the MOEAs (shown as black circles) as well as the probabilities of successfully attaining a given percent of the best values for the generational distance, normalized ε-indicator, and hypervolume metrics (shown with gray shading).

distance LTM plot in Fig. 7b). When interpreting Fig. 7, if the black circles designating the best performance of an MOEA fall below 100% of the best metric value, then it means that an algorithm has no sampled run that attained ideal performance. Also if the attainment probabilities fall below the black circles designating best performance, then the MOEA is highly sensitive to its parameterizations and it cannot reliably attain its best performance.

The generational distance results in Fig. 7 confirm that the LRGV water management problem is the most challenging of the three applications. Several of the algorithms' best performances were not close to their ideal value for the LRGV problem. The NSGAII and the GDE3 fail to reliably attain generational distance results within 75% of the best metric value. Given that generational distance is the easiest of the metrics to satisfy, these results represent a severe failure for these algorithms. It should also be noted MOEA/D shows a severe failure for the HBV test case in terms of its generational distance results. Recall from Section 3.4 that MOEA/D uses Chebyshev-based aggregation of objectives into a single objective weighted function. The HBV results show that MOEA/D can be severely impacted by poor relative scaling across objectives. The ε-indicator results further distinguish performance across the algorithms in terms of their consistency (i.e., representing solution in all regions of the reference set as illustrated in Fig. 2c).

In the HBV calibration, it is interesting to note that although the ε-MOEA was able to attain the best overall ε-indicator metric value from a single run, this level of performance is highly unlikely. Except for MOEA/D, the ε-MOEA has one of lowest probabilities of success. This reflects the ε-MOEA's strong sensitivity to its parameterization, making it very difficult to use successfully on the HBV calibration application. The hypervolume results for the HBV application in Fig. 7 are even more variable, especially with respect to the probability of attainment. The Borg, GDE3, IBEA, and ε-NSGAII are the top overall performers in terms of their best performance and maintaining a high attainment probability very near their best performance. For these top performing algorithms, the close proximity of their best performance to the best metric as well as their high attainment probabilities signifies that any single run would have a very high quality approximation to the best known Pareto front. Alternatively, MOEA/D, AMALGAM, ε-MOEA, OMOPSO, and NSGAII suffer from reduced hypervolume attainment probabilities for the HBV calibration.

Fig. 7b and c show that the best ε-indicator and hypervolume results for the LTM and the LRGV applications are far below their best metric values (typically 40–75% of the ideal). This is a direct contrast to the HBV calibration results where a single algorithm run can attain a high quality representation of the Pareto front (in terms of convergence and diversity). For the LTM and LRGV cases, it is necessary to run even the most effective parameterizations for multiple seeds to attain full representations of their Pareto frontiers, as has been shown in the prior published results for these test cases [18,83,39]. Multiple random seeds contribute diversity to a given MOEA's search results. Recall from Section 5.3 that we expected all of the MOEAs tested in this study to struggle to converge a highly diverse approximation to the LTM problem's reference set. The LTM test case's decisions have complex spatial interdependencies that emerge based on how groups of sampling points impact plume characterization objectives. The results in Fig. 7b confirm our prior results that the interdependencies in the LTM test case's decisions are not adequately captured with traditional recombination and mutation operators (i.e., all of the algorithms are suffering from epistasis failures – see Kollat et al. [83] for more details). It is also worth noting that although most of the MOEAs tested were able to find portions of the LTM reference set (see Fig. 6), the problem is severely difficult with respect to finding well converged, consistent, and diverse solution sets. Overall MOEA/D, IBEA, and AMALGAM were the weakest algorithms for the LTM case. The remaining algorithms all had fairly consistent best run and attainment results.

The LRGV's best run and attainment for the ε-indicator and hypervolume metrics are highly variable across the algorithms in Fig. 7c. This problem caused the strongest mismatch between the best single run and the attainment probabilities (i.e., the likelihood of success). Overall the OMOPSO was the best performer for this problem for the highest levels of attainment. Borg and OMOPSO had the best performing single runs. The significant success of the OMOPSO is somewhat surprising and would be difficult to predict a priori. The OMOPSO's performance in the DTLZ2, HBV, and LTM test cases was not strongly distinguished. Overall, OMOPSO, ε-NSGAII, and Borg were the top performing algorithms for the LRGV test case. The NSGAII, the SPEA2, and the IBEA had the worst performance for the problem. Interestingly, Fig. 7 highlights that the GDE3 had a high probability of success for the 50% threshold for ε-indicator but virtually a zero probability of any hypervolume. Recall in Fig. 5c that the LRGV test case has two distinct regions in its best known Pareto front. The GDE3 does a good job of finding solutions that represent each of these regions but does a very poor job converging these solutions toward their best values (i.e., the algorithm prematurely converges).

Figs. 8–10 help to provide a more detailed understanding of the best and attainment results by presenting detailed control maps for the ten tested MOEAs across the HBV, LTM, and LRGV test cases, respectively. These figures show color control maps of hypervolume performance as a function of population size and the NFE used in search. The control maps provide a sense of the "sweet spot" [64] for algorithms' parameter spaces (i.e., how sensitive an algorithm is to its parameters). Ideal performance would be a plot that is fully dark blue, representing an algorithm that achieves the highest percent of the theoretical hypervolume attained in this study regardless of how it is parameterized (i.e., an algorithm where it is difficult to cause search failures due to its parameterization). The color legends in Figs. 8–10 provide a quantitative measure of the percent of ideal hypervolume captured across 50 random seeds. As discussed in Section 4.1, we utilize control maps plotting population size versus the NFE used since these parameter choices are dominant drivers for MOEA performance and computational constraints. Note that in Figs. 8–10, we are emphasizing the ranges of hypervolume attained by the best MOEAs to clarify their performance differences. This is in contrast to Fig. 7 which emphasizes best and attainment performance relative to the theoretical maximum hypervolumes. Another difference between the control maps and the attainment results is that Fig. 7 focuses on the probabilities of success assuming a single random seed trial. The control maps in Figs. 8–10 alternatively evaluate the MOEAs' hypervolumes attained across 50 random seed trials.

Fig. 8 confirms the HBV hypervolume attainment results from Fig. 7. The color legend in Fig. 8 shows that peak hypervolume performance (designated as dark blue) is very near 90% of the ideal value. The Borg HBV control map shows that the algorithm has a very broad range of effective parameters for its hypervolume performance. The algorithm is more effective relative to the other MOEAs for lower population sizes and relatively small NFE. The ε-NSGAII also shows a broad zone of high performance, although it is generally inferior to Borg for lower NFE. Both Borg and ε-NSGAII exploit adaptive population sizing, which as shown in Fig. 8 helps them to perform well independent of their initial population sizes. They both benefit from smaller initial populations that facilitate an increased number of generations of evolution. Although IBEA shows a strong control map in Fig. 8, its performance is more complex where small population sizes and increased NFE actually degrade its performance. MOEAs with complex control maps are difficult to use, especially if they have islands of high performance that are difficult to predict. GDE3 and AMALGAM consistently require

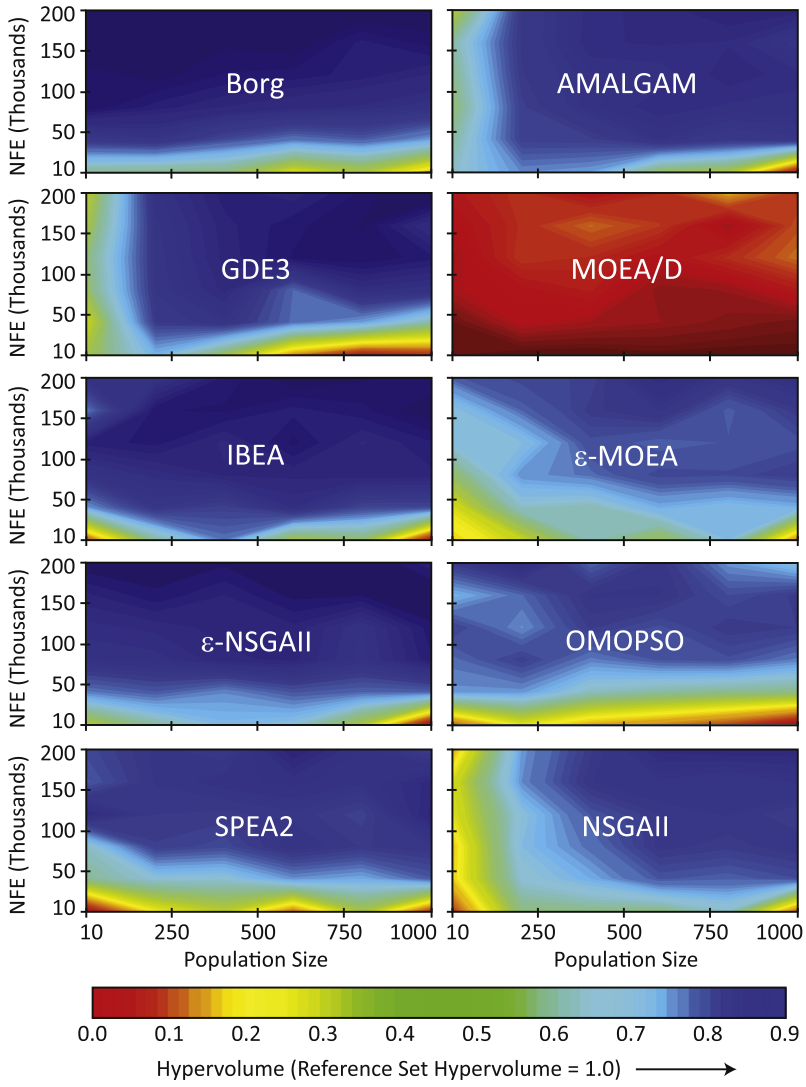


Fig. 8. Hypervolume performance control maps for the HBV test case. The color scale represents how performance varies for increasing population sizes and function evaluation counts. Ideal performance is shown in zones of dark blue shading while severely poor performance is designated by dark red. The control maps are based on the full set of Latin hypercube samples of the parameters for each algorithm. The hypervolumes were computed relative to the approximation sets attained across the 50 random seed runs used to evaluate each parameterization.

larger population sizes and increased NFE to improve their performance. In many instances of Fig. 8, the HBV calibration control maps are more complex where distinct patterns are hard to predict and consequently effective hypervolume performance would be difficult to capture in applications. For example, although SPEA2 and OMOPSO have zones of high performance, they do not have clear trends, yielding isolated parametric islands of good performance. Surprisingly, MOEA/D fully fails on the HBV test case as was noted in our prior discussion of its attainment results in Fig. 7a. Although this algorithm has been shown to be very strong in solving test functions, its sensitivity to the relative scaling of an application's objectives makes it more difficult to predict if the algorithm will be successful in water resources applications.

The control maps for the LTM application in Fig. 9 designate hypervolumes between 35% and 70% of the ideal value. As was noted in Fig. 7b, all of the algorithms failed to attain highly converged, diverse representations of the LTM application. The control maps in Fig. 9 show a high degree of variability across the algorithms. Confirming the attainment results in Fig. 7b, both Borg and GDE3 have the best overall control maps. Only Borg has high levels of performance for reduced population sizes and reduced NFE. AMALGAM also performs reasonably well for the LTM test case in Fig. 9, clearly benefitting from the use of 50 random seed trials. AMALGAM shows a peak of performance near a population size of 500 and a maximum NFE within a broader region of improved performance. Generally for the LTM application, even though several of the algorithms have zones of high performance, peak performance would be difficult to attain in practice without some form of *a priori* global sampling of their parameterizations. Islands of high performance are highly challenging and designate that the algorithms are not "controllable" (i.e., the "needle-the-haystack" parameterization challenge). Users would struggle to know *a priori* how to capture their peak performance.

Fig. 9 highlights the fallacy that MOEA developers know the parameterizations that attain peak performance across applications. A key insight from Fig. 9 is that for the highly controllable algorithms for the LTM test case (i.e., Borg and GDE3), the use of default parameterizations are better justified because it is generally difficult to make the algorithms fail given sufficient NFE. That being said, GDE3 does have a strong sensitivity to its population size and its peak performance is contingent on the user specifying a population size greater than 100. In contrast, Borg's auto-adaptive

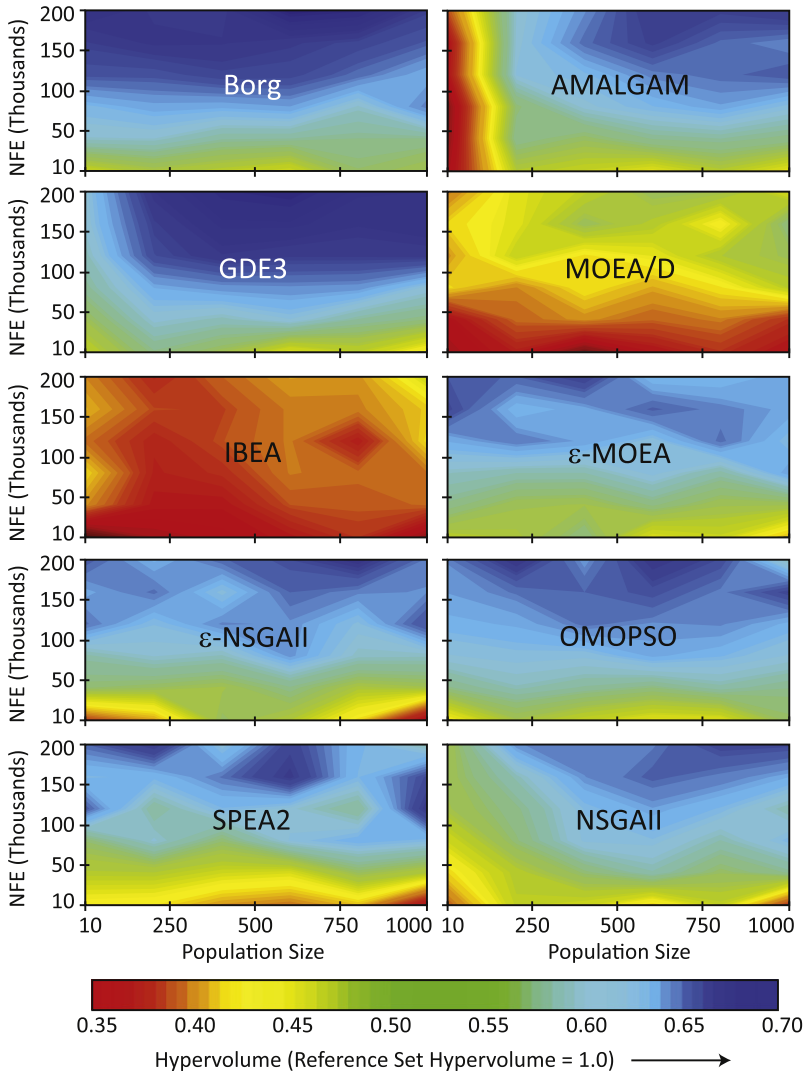


Fig. 9. Hypervolume performance control maps for the LTM test case. The color scale represents how performance varies for increasing population sizes and function evaluation counts. Ideal performance is shown in zones of dark blue shading while severely poor performance is designated by dark red. The control maps are based on the full set of Latin hypercube samples of the parameters for each algorithm. The hypervolumes were computed relative to the approximation sets attained across the 50 random seed runs used to evaluate each parameterization.

population sizing makes its search robust regardless of the initial population. Borg's performance is controllable by simply increasing the NFE available to search. Another very interesting overall result in Fig. 9 is that Borg, GDE3, and AMALGAM all have rotationally invariant recombination operators that can capture a modest degree of decision variable interactions (i.e., epistasis). Although Fig. 7b still highlights that all of the MOEAs suffered from epistasis failures, Fig. 9 highlights that rotationally invariant search operators did aid in improving the effectiveness and controllability of search (as did the consideration of 50 random seed trials).

Fig. 10 highlights that the highly constrained and stochastic LRGV test case is challenging for all of the MOEAs. As was noted by Tang et al. [89] MOEA search failures should be distinguished between two cases: (1) they need more NFE to make sufficient progress to solve a problem or (2) regardless of NFE or other parameter choices the algorithm is incapable of solving a problem. Fig. 10 highlights both of these failure modes. The top performing algorithms for the LRGV test case are Borg, OMOPSO, ε-NSGAII, and ε-MOEA. It is interesting to note that all of these algorithms use ε-dominance archiving and that they suffer from the first failure mode (i.e., they need higher NFE values for success on the LRGV test case). The two top performers on the LRGV test case are Borg and OMOPSO. Recall in Fig. 7c that OMOPSO had the highest attainment probabilities for a single random seed trial while Borg had the best overall hypervolume. Fig. 10 further explains these observations and demonstrates the benefits of using 50-random seed trials for each LHS parameterization. At maximum NFE, Borg clearly attains very near peak performance across the top of its control map. At lower NFE, Borg requires a smaller initial population size to have sufficient evolution to attain high quality Pareto approximate results. Borg's attainment probabilities are degraded by the zone of reduced performance occurring where its initial population size ranged between 500 and 1000 because the algorithm did not have a sufficient number of generations of evolution. Alternatively, OMOPSO never attains the highest hypervolumes but it is more consistent across the range of tested population sizes given sufficient NFE. Figs. 8–10 generally show that Borg performs ideally when its initial population size is 100 and given sufficient NFE. Fig. 10 shows that AMALGAM and MOEA/D are second tier performers that also appear to be suffering from the first failure mode (i.e., insufficient NFE). Alternatively, SPEA2, NSGAII, and IBEA appear to be fully failing across all tested parameterizations (i.e.,

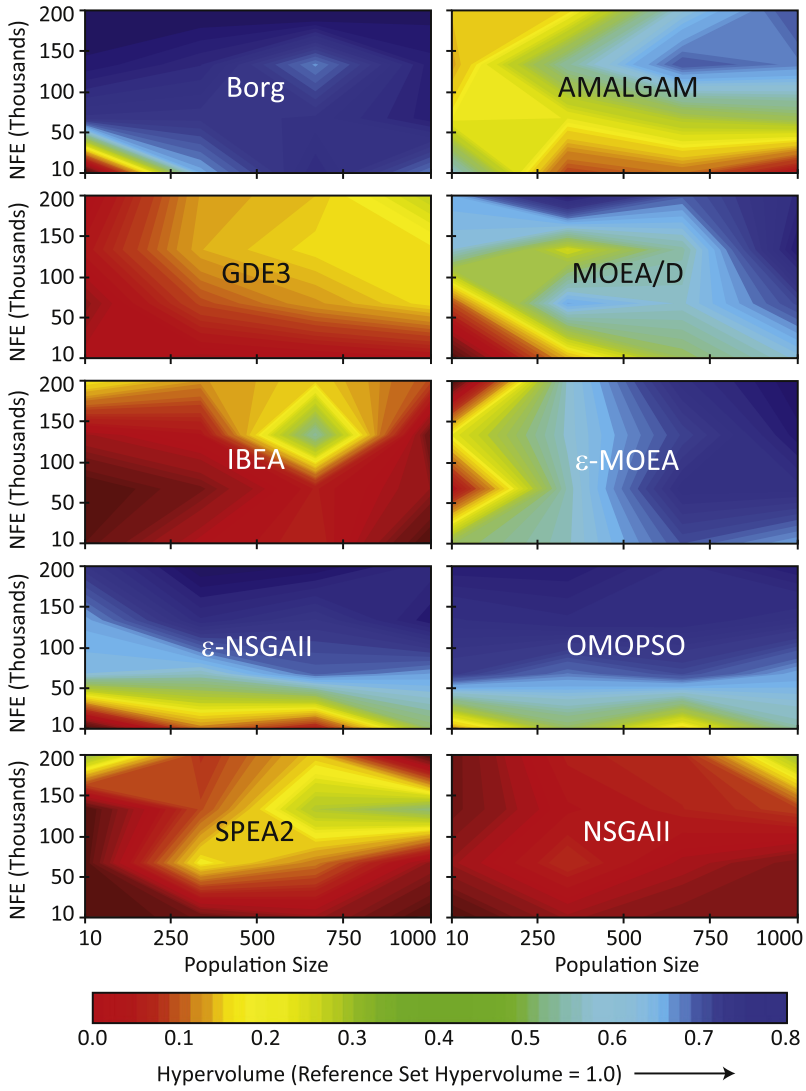


Fig. 10. Hypervolume performance control maps for the LRGV test case. The color scale represents how performance varies for increasing population sizes and function evaluation counts. Ideal performance is shown in zones of dark blue shading while severely poor performance is designated by dark red. The control maps are based on the full set of Latin hypercube samples of the parameters for each algorithm. The hypervolumes were computed relative to the approximation sets attained across the 50 random seed runs used to evaluate each parameterization.

the second type of failure). IBEA provides an another interesting example of the "needle-in-the-haystack" parameterization challenge. Fig. 10 shows that IBEA has a small island of slightly improved performance (shown in green) that could not be easily predicted a priori.

A key concern that arises from Figs. 5–10 is the significant difficulty that users face in predicting which MOEA to use for a given application. Although the OMOPSO has very pronounced performance for the LRGV test case, it had reduced scalability for the DTLZ2 test problem and limited hypervolume attainment for the HBV calibration problem (see Figs. 4 and 7). Additionally, this study as well as prior work [36,16] shows that increases in objective count can cause the "sweet spots" for MOEAs' parameter spaces to vary dramatically, making it exceedingly difficult to effectively parameterize search. Vrugt and Robinson [71] correctly highlight that across different problems, the benefits and limitations of different MOEA search operators change causing standard non-adaptive MOEAs to struggle. Both Borg and AMALGAM do have the benefit of adapting their search operators as part of their evolutionary process. Moreover, Borg and the ε-NSGAII share adaptive population sizing to automatically adjust their search populations commensurate with the Pareto approximation sets they find. Figs. 5–10 demonstrate that only Borg has consistently provided either the top performance or competitive performance across all of the problems in terms of its scalability, capturing reference sets, best performance, probabilistic attainment, and controllability.

7. Conclusions

The water resources field is being challenged to better define the scientific basis by which we can promote sustainability, adaptivity, and reliability in our water resources systems while acknowledging their nonstationarity. Although these challenges are not new as reflected by the pioneering early work in multiobjective optimization within our field [3,1], breakthroughs in evolutionary multiobjective optimization can reshape our ability to discover the many-objective observation, prediction, and management tradeoffs that will shape the future of our water resources systems. This study utilized more than 100,000 separate MOEA runs and trillions of function evaluations to carefully evaluate 10 state-of-the-art MOEAs. Our results provide a clear state-of-the-field assessment for many-objective optimization within water

resources applications. Given the breadth of the results, problem types, and algorithms tested in this study, a very brief synopsis of the relative strengths and weaknesses of the top performing, benchmark MOEAs are provided in this section. Our insights into these benchmark algorithms draw specifically from this study and a broader diagnostic assessment of MOEAs across a diverse suite of problem types beyond those tested here (see Hadka and Reed [16]).

Borg. The Borg algorithm showed the best scalability on the DTLZ2 test problem with objective counts as high as 16. It was also a dominant contributor to all of the applications' best known Pareto fronts. The algorithm exploits adaptive operator selection and adaptive population sizing to attain both high probabilities of success (''attainment'') and controllability (''sweet spots'' on control maps). Generally, Borg was the only algorithm to consistently exploit small populations and reduced NFE to attain competitive to superior ε-indicator and hypervolume results. Borg's control maps show that the algorithm's performance is highly controllable and quickly improves with increases in NFE (experimentation with its other parameters would be unnecessary). Overall Borg represents a top performing MOEA that could strongly support many-objective water resources applications. It was the most consistent of the ten algorithms tested across the problem suite. Borg also satisfies the requirements for having both a theoretical proof of convergence and diversity maintenance.

GDE3. GDE3 exploits the differential evolution search operators and is consequently capable of rotationally invariant search (i.e., use on non-separable real-valued problems). It is one of the most parsimonious MOEAs tested in this study. It exhibited a somewhat reduced scalability for large objective counts on the DTLZ2 test case and did not contribute to the HBV or LRGV best known reference Pareto fronts. The algorithm did have high success rates (''attainment'') on the HBV calibration and the LTM problems. In the highly constrained and stochastic LRGV test case, the algorithm suffered from premature convergence and poor hypervolume attainment. Although GDE3 has a limited number of parameters, its control maps highlight that the algorithm necessitates careful experimentation to ensure an effective parameterization (especially with respect to its population size and NFE).

ε-NSGAII. The ε-NSGAII was the first algorithm to utilize adaptive population sizing and ε-dominance archiving. Borg improves upon many of these innovations in its search. Overall ε-NSGAII was one of the most scalable algorithms and was a major contributor to the LTM test case's best known Pareto front. It was also a top performer in terms of its ε-indicator and hypervolume success probability attainments for the HBV calibration. The ε-NSGAII showed a reduced controllability and sensitivity to its initial population size for the LTM test case. Its controllability for the HBV and LRGV test cases was competitive with other top-performing algorithms, although it generally required larger initial search populations and a higher NFE. Overall ε-NSGAII performed well but its performance would be expected to be exceeded by Borg. The ε-NSGAII does satisfy the conditions necessary for a theoretical proof of convergence and diversity.

AMALGAM. AMALGAM warrants recognition as the first auto-adaptive multi-operator MOEA. The algorithm struggled in its scalability and contributions to the water resources applications' reference sets. Although AMALGAM did not have peak performance, it did perform well for the HBV and LTM test cases. Its control maps do show a strong sensitivity to its parameterization (i.e., less controllability), where generally it required higher population sizes and more NFE relative to the top performing MOEAs. AMALGAM struggled on the stochastic and highly constrained LRGV test case. As with ε-NSGAII, AMALGAM's performance would generally be expected to be exceeded by Borg.

OMOPSO. OMOPSO is unique among the algorithms as the only representative of the particle swarm heuristics. The algorithm has been shown to be promising on a range of problems in the evolutionary computation literature and benefits from exploiting an ε-dominance archiving strategy. OMOPSO had a reduced performance in terms of scalability for the DTLZ2 test problem relative to the other algorithms that include ε-dominance (Borg, ε-NSGAII, and ε-MOEA). It also had limited contributions to the best known Pareto fronts for the three water resources applications. OMOPSO had variable attainment success rate probabilities across the applications. It struggled to reliably attain high hypervolumes for the HBV calibration. OMOPSO was more consistent on the highly constrained, stochastic LRGV test case in terms of both attainment and controllability. OMOPSO satisfies the conditions necessary for a theoretical proof of convergence and diversity.

Overall the Borg, GDE3, ε-NSGAII, AMALGAM, and OMOPSO MOEAs represent the top performing modern tools. The Borg MOEA was the only algorithm to perform consistently at competitive to superior performance levels on all of the applications. Borg also shows the highest controllability (or ease-of-use) of all of the tested MOEAs. Overall the top performing algorithms in this study provide a strong baseline for future research advancing the size and scope of the many-objective water resources applications that can be considered. These algorithms should be the focus for future benchmarking studies, which need to follow rigorously constructed statistical experiments such as the one contributed in this study.

This paper also highlights key issues that should be considered for future MOEA research contributions in the water resources literature. For MOEAs to fully realize their potential for operational use in water resources systems decision making, they need to be controllable, efficient, and reliable across a broad suite of problem types. As initially highlighted by Vrugt and Robinson [71] when introducing AMALGAM, and reinforced with the success of the Borg MOEA in this study, we need to advance auto-adaptive algorithm frameworks that tailor their search according to a given problem's traits with minimal user input. We need to refocus our literature away from studies that seek to find which problem-specific MOEA parameters work. Instead we should focus on developing new auto-adaptive MOEAs that make it difficult to find parameterizations that fail. As we transition to these auto-adaptive tools, this special issue provides an ideal reminder of how the innovations over the past 35 years have dramatically reshaped our capabilities to employ multiobjective optimization in water resources. As we transition toward a future of many-objective optimization, our field needs to broadly explore advances in human–computer interaction, the emergence of ubiquitous parallel cloud computing services, and visual analytics. These complementary fields can accelerate our discovery of the water resources planning innovations needed for ensuring a sustainable future.

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